

CAPSTONE PROJECT PRESENTATION

Resume Screening & Role Matching using NLP

An Enterprise AI-Powered Applicant Tracking System (ATS)

COMPANY

TalentBridge Solutions Pvt. Ltd.

PREPARED BY

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COURSE

Data Science with GenAI

REGISTRATION ID

The Business Problem

1

Manual Screening

Recruiters spend hours reading resumes one by one.

2

Keyword-only ATS

Traditional ATS tools miss contextually skilled candidates.

3

Inconsistent Decisions

Different recruiters judge the same resume differently.

4

High Recruitment Volume

Thousands of resumes per role delay the hiring process.

THE NEED FOR AUTOMATION

TalentBridge Solutions needs an intelligent, consistent, and explainable way to screen resumes at scale.

Project Objective

Why this project was built:

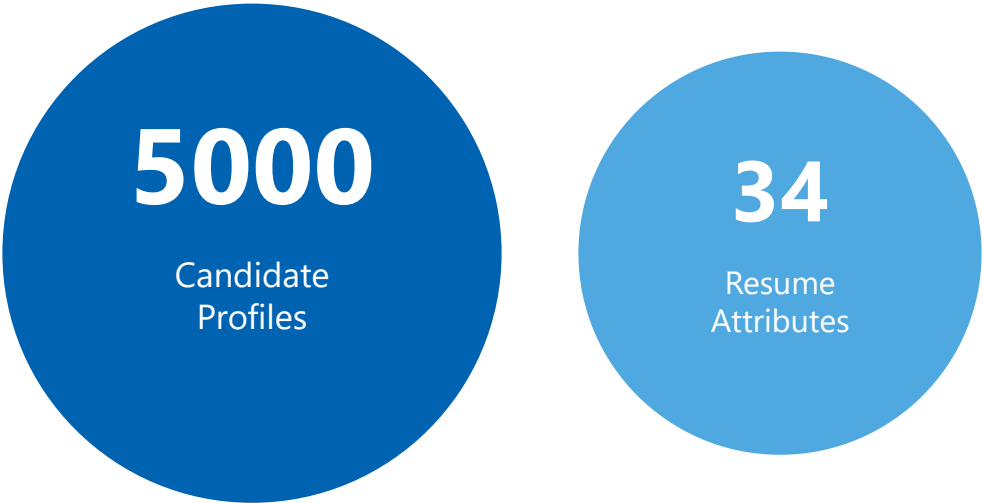
- ✓ Automate resume screening for enterprise roles
- ✓ Improve candidate-to-role matching using NLP
- ✓ Reduce recruiter workload and manual effort
- ✓ Generate explainable ATS Fit Scores (0-100)
- ✓ Provide an interactive recruiter dashboard

EXPECTED OUTCOME

A faster, fairer, and more consistent hiring process that helps recruiters focus only on the most relevant candidates.

Dataset Overview

`parsed_resumes.csv`

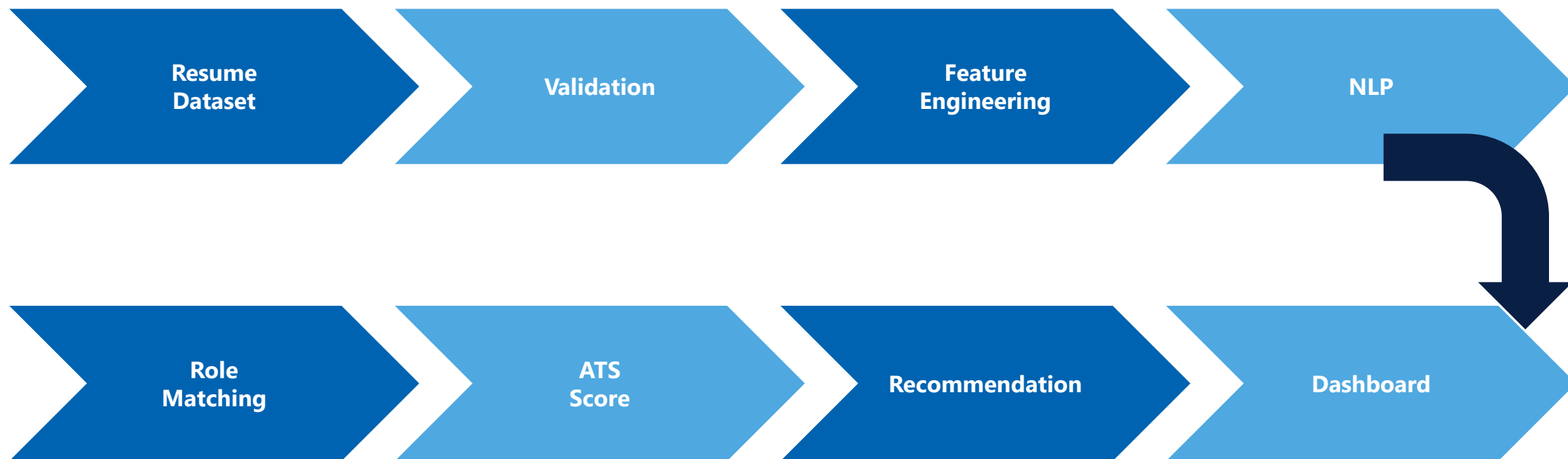


Key Fields Captured Per Candidate

Technical Skills	Tools & Platforms
Experience Summary	Education Details
Project Summary	Leadership Flags
ATS Quality Scores	Domain & Role

Data quality validated before modeling: 0 missing values | 0 duplicate candidate IDs | fields standardized (education, skills)

Project Workflow



A structured pipeline that converts raw resumes into ranked, explainable recruiter recommendations.

Feature Engineering

1

Resume Quality Score

How complete and well-written the resume is.

2

ATS Readiness Score

How compatible the resume is with ATS systems.

3

Technical Strength

Depth of technical skills, tools and cloud experience.

4

Leadership Score

Management, mentoring and stakeholder skills.

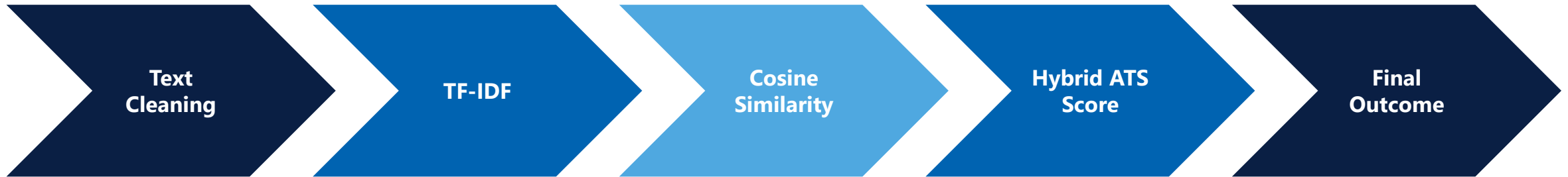
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Candidate Quality Index

One overall score combining all key features.

Why: Raw resume text alone cannot show candidate quality. These engineered features turn resumes into measurable, comparable business indicators.

NLP & ATS Scoring Workflow

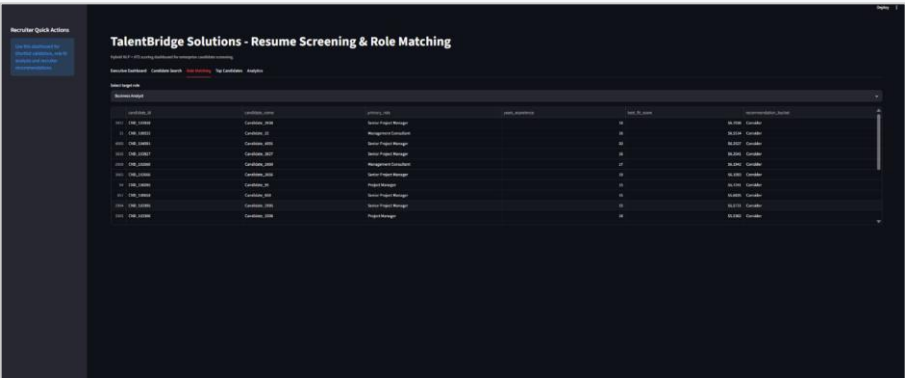


In simple words:

- Resume text is cleaned and simplified (removing filler words).
- TF-IDF turns text into numbers, highlighting important skill words.
- Cosine Similarity checks how closely a resume matches a job role.
- This is combined with experience, education and leadership into one Hybrid ATS Score.
- The score decides the final recruiter recommendation.

SEE IT IN ACTION

Live Project Demonstration



TalentBridge Solutions - Resume Screening & Role Matching

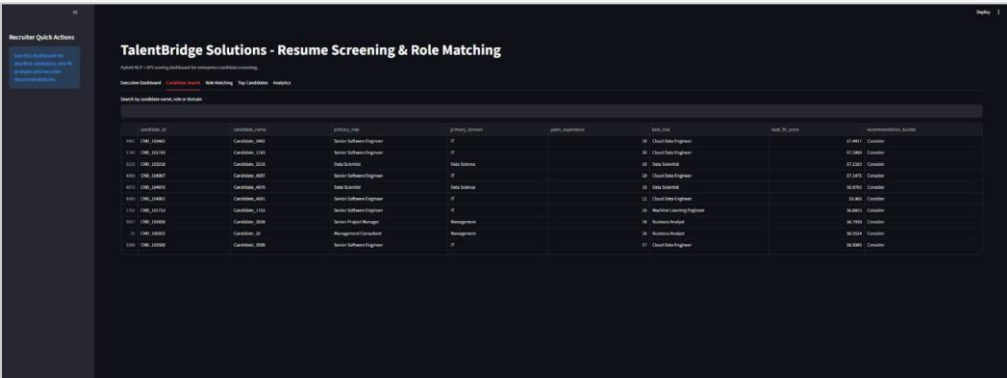
Recruiter Quick Actions

Search by: All Matching Candidates | Candidates | Top Candidates | Analytics

Search by candidate name, role or domain

Candidate ID	Candidate Name	Primary Job	Primary Location	Years Experience	Score	Recommendation Score
101	John Doe	Software Engineer	IT	5	95	90
102	Jane Smith	Product Manager	IT	3	88	85
103	Mike Johnson	Software Engineer	IT	7	92	88
104	Sarah Lee	Software Engineer	IT	4	85	82
105	David Kim	Software Engineer	IT	6	90	87
106	Emily White	Software Engineer	IT	2	80	78
107	Chris Brown	Software Engineer	IT	8	98	95
108	Alex Green	Software Engineer	IT	1	75	72
109	Mia Black	Software Engineer	IT	9	99	96
110	Noah Grey	Software Engineer	IT	0	70	68

Role Matching



TalentBridge Solutions - Resume Screening & Role Matching

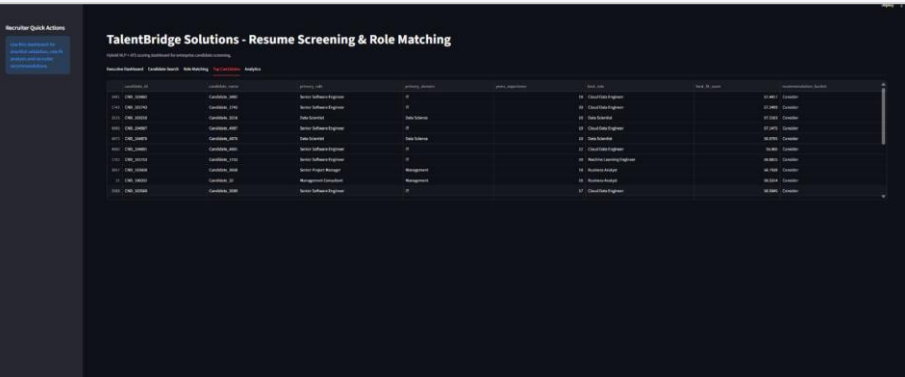
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Candidate Search



TalentBridge Solutions - Resume Screening & Role Matching

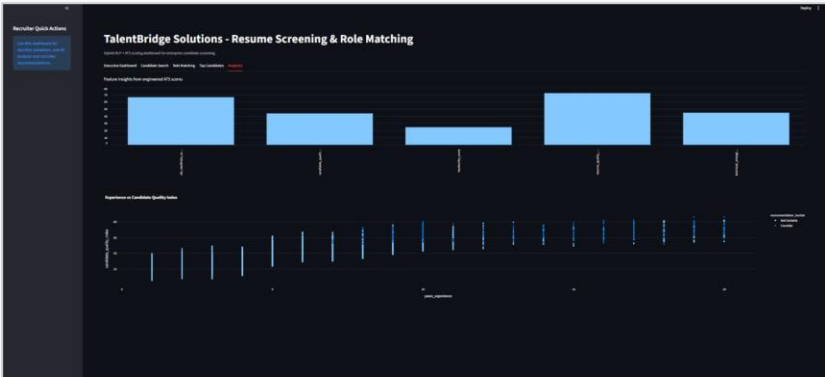
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Top Candidates



Analytics

Results & Business Insights

Project Achievements

- Processed 5,000 candidate profiles end-to-end
- Engineered 15+ business-driven features
- Matched candidates across 5 enterprise roles
- Automated, explainable ATS Fit Scores (0-100)
- Delivered an interactive recruiter dashboard

KEY OBSERVATION FROM ANALYTICS

Average Fit Score across all candidates: ~36.6 / 100

Highest Fit Score achieved: ~57.4 / 100

Most candidates fall in the 'Consider' or 'Not Suitable' bands rather than 'Highly Recommended'. This is expected and realistic: the dataset spans many domains (IT, Legal, Management, Data Science), while only 5 specialised enterprise roles were defined. The scoring engine correctly avoids inflating scores for candidates outside these target roles - proving the model is honest and business-realistic, not just optimistic.

Future Enhancements

1

Resume PDF Parsing

Accept real PDF/DOC resumes directly.

2

BERT / Transformers

Deeper semantic understanding of text.

3

Large Language Models

GenAI-based resume summarisation & scoring.

4

Cloud Deployment

Host the solution for real recruiter access.

5

Explainable AI (XAI)

Visual reasons behind every candidate score.

6

Real ATS Integration

Connect with live hiring platforms.

7

Recruiter Feedback Learning

Improve scoring using recruiter decisions over time.

CONCLUSION

**A complete, explainable ATS prototype -
from raw resumes to recruiter-ready recommendations.**

- Combines NLP similarity with real business scoring factors
- Turns 5,000 resumes into ranked, explainable recommendations
- Delivered through a practical, recruiter-friendly dashboard

Thank You

Questions?